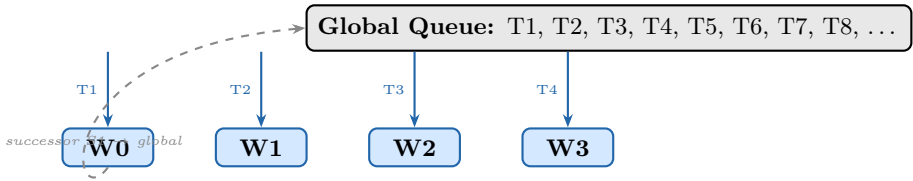
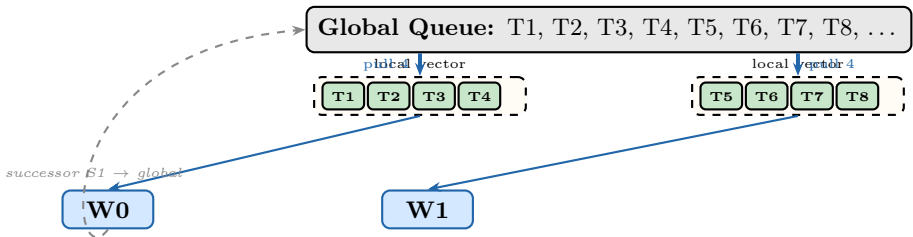


GLOBAL_QUEUE



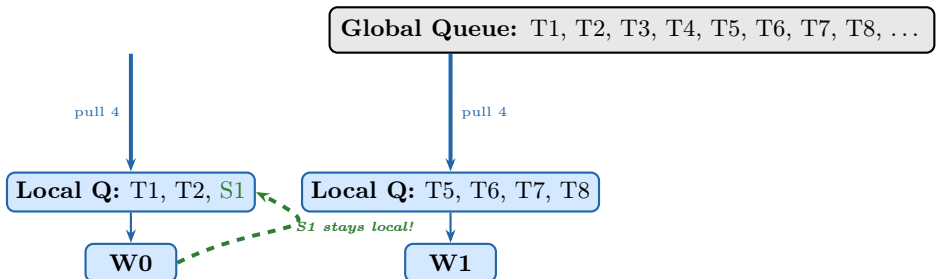
- Each worker pulls **one task** at a time
- Each pull/push = one atomic operation on shared queue
- Successors go **back to global queue**
- Simple, fair, but high contention at scale

BATCH_PULL (batch=4)



- Workers pull **N tasks at once** (1 blocking + N-1 non-blocking)
- Process from local vector until empty, then refill
- Successors go **back to global queue**
- N× fewer atomic ops on shared queue

HYBRID_QUEUE (batch=4)



- Workers pull **N tasks at once** from global into **per-thread queue**
- Drain local queue first (successors + pulled tasks)
- Successors **stay on local queue** → zero contention
- **Cache locality:** successor data still hot in L1/L2
- Refill from global only when local queue is empty